

EE3621: Digital Signal Processing

Lecture 25: FIR Filters - moving average filters (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Analyze** the frequency selectivity and pole-zero placement of Moving Average filters.
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2. MATHEMATICAL FOUNDATIONS

2.1 Moving Average Filter

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] \implies H(z) = \frac{1-z^{-M}}{M(1-z^{-1})}$$

$$H(e^{j\omega}) = \frac{1}{M} \frac{\sin(\omega M/2)}{\sin(\omega/2)} e^{-j\omega(M-1)/2}$$

- $M - 1$ zeros are located on the unit circle at $z = e^{j2\pi k/M}$ for $k = 1, 2, \dots, M - 1$.
- The pole at $z = 1$ exactly cancels the zero at $z = 1$.

Pictorial Representations

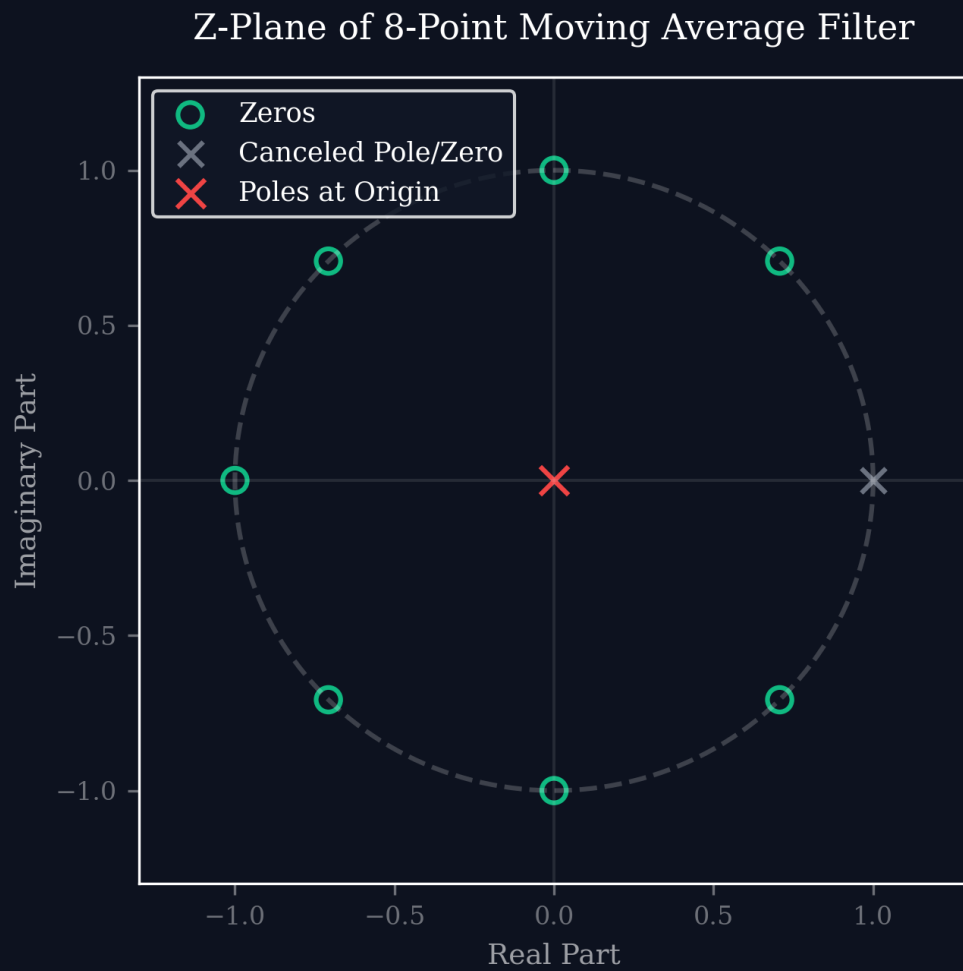


Figure 1: Moving Average Zplane

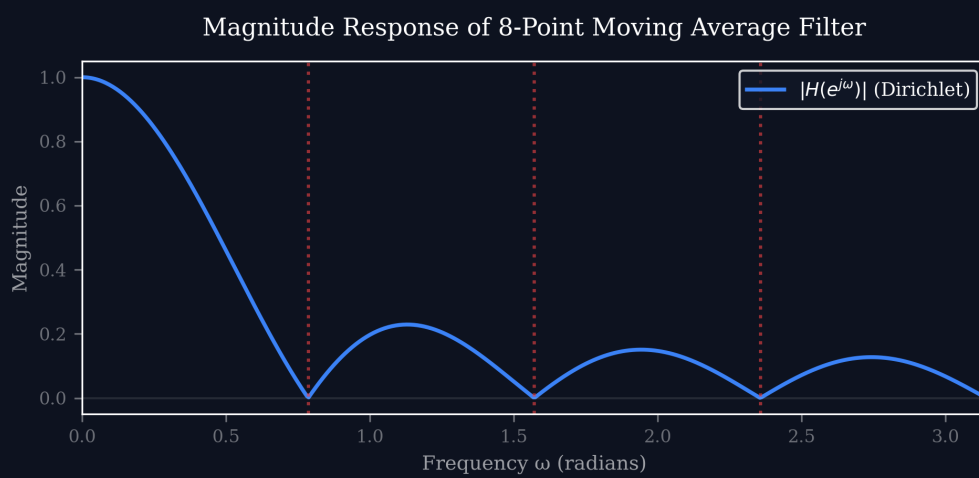


Figure 2: Moving Average Frequency